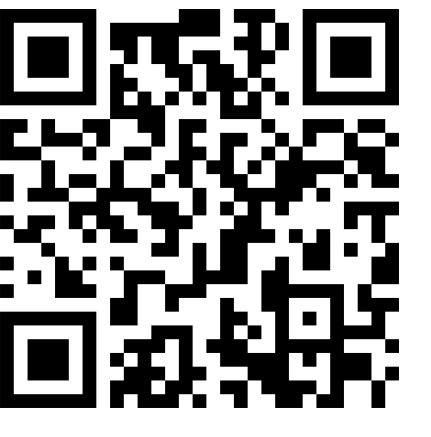




Beyond Faces: The Effect of Contexts on Ensemble Emotion Perception

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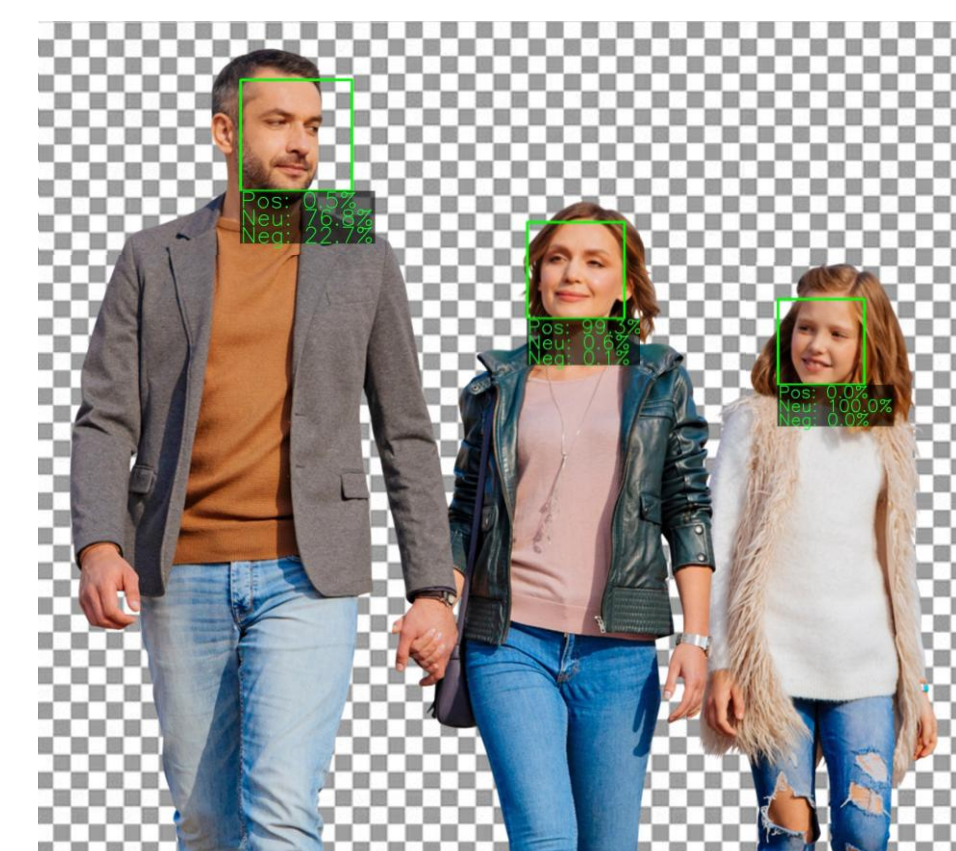
Background

- People are good at judging the **overall mood of a crowd** based on **facial expressions** (Son et al., 2023)
- Background context** influences crowd emotion judgments (Hao et al., 2024), but the effect of situational context, separate from facial and bodily cues, remains unclear
- Observer's emotional state** may also influence ensemble emotion perception, as positive mood broadens attentional scope, facilitating the integration of information (Peng et al., 2021; Baek & Chong, 2020)

Question

How do facial expressions, background context, and the observer's emotional state jointly influence ensemble emotion perception?

Stimuli

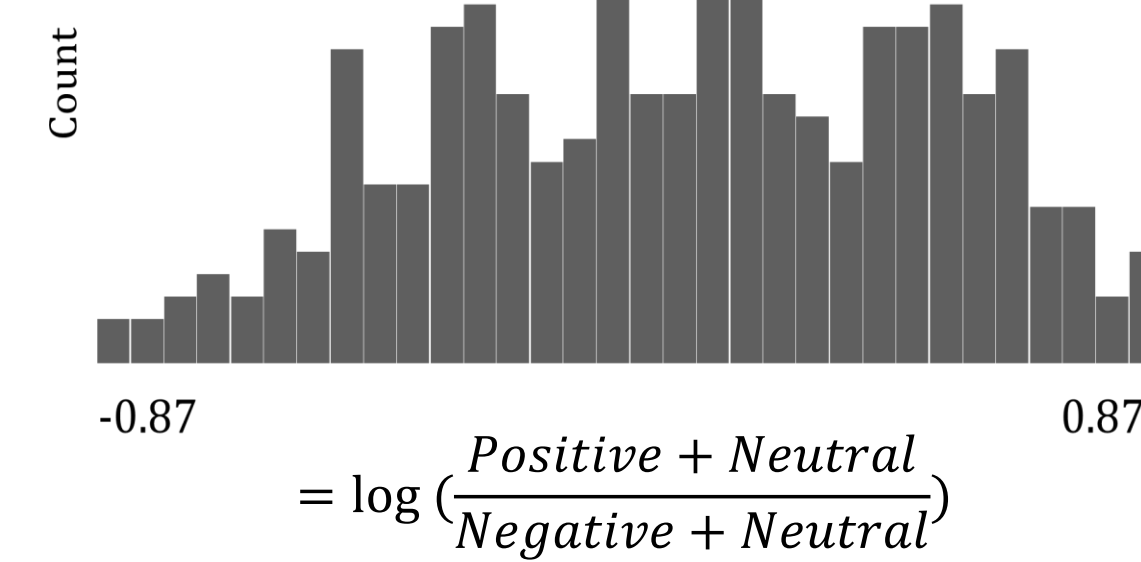


319 crowd images

DeepFace

: 7 emotions → 3 groups
(Positive, Neutral, Negative)

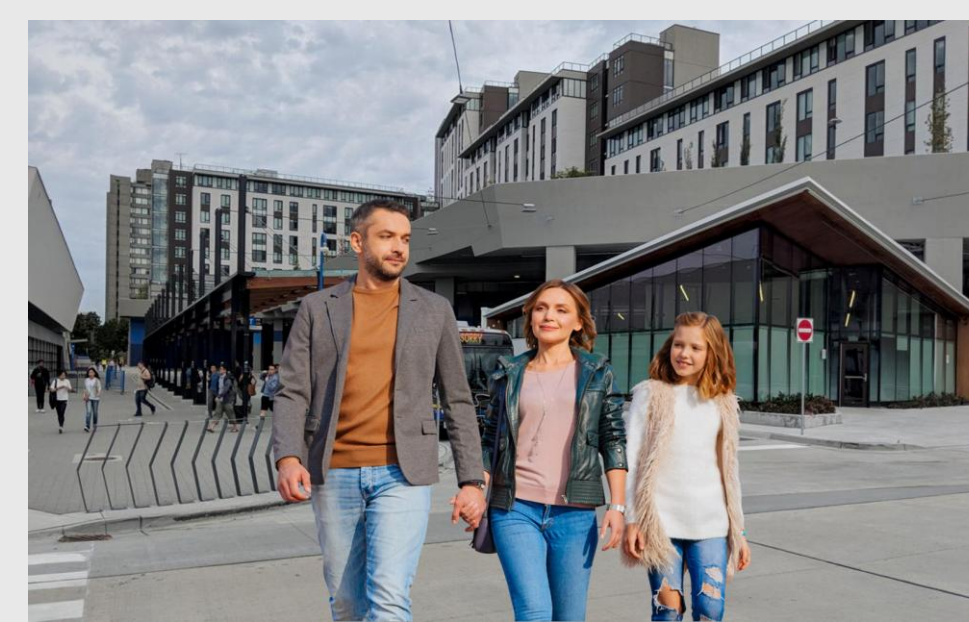
Facial Emotion Value



Positive background



Neutral background



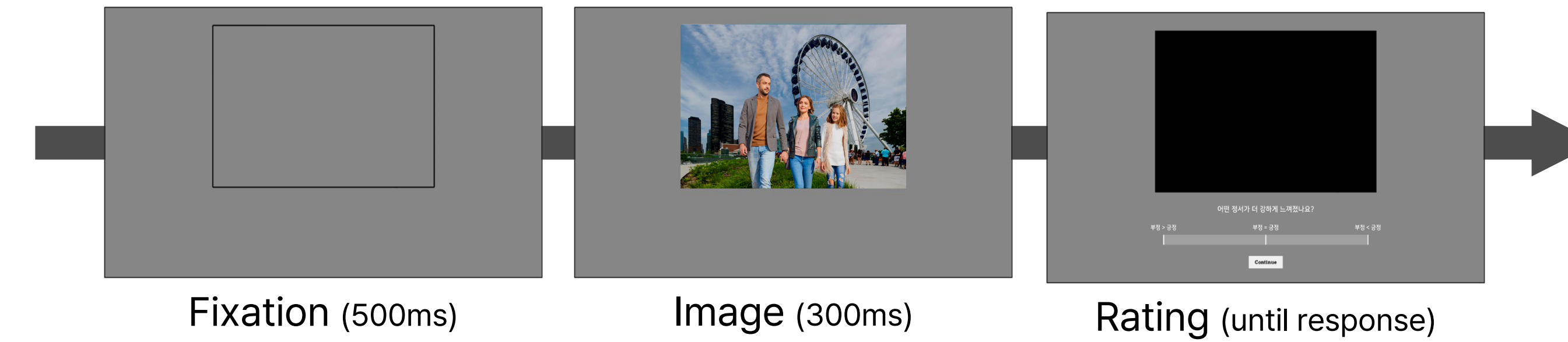
Negative background



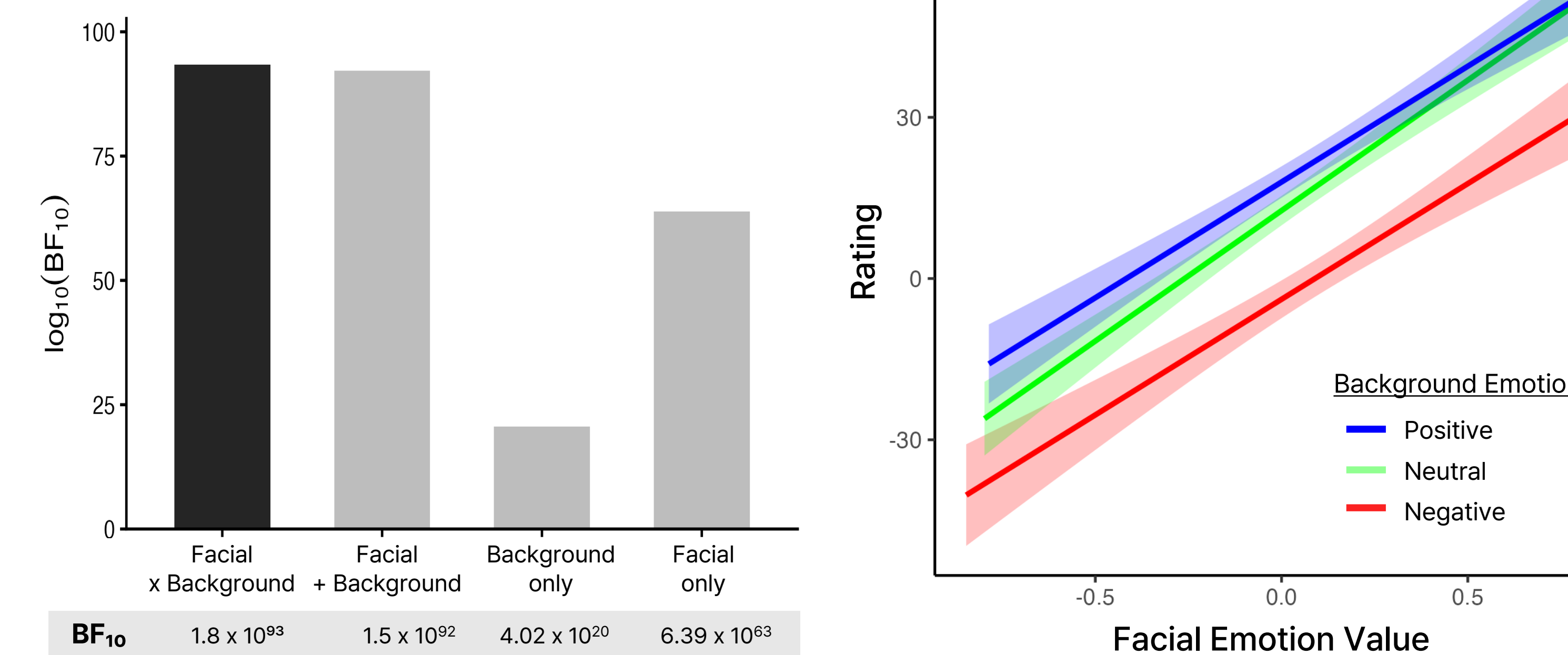
- Background emotion**: rated independently (N = 14) → 20 images per valence
- Crowd images** were randomly composited onto background images

Experiment 1

- Emotion ensemble task**: report the average emotion of the crowd scene



Results (N = 10)



- The **full model** (facial emotion × background emotion) provided the **best fit** relative to the null model ($BF_{10} > 10^{93}$)
- Background emotion biases ensemble judgments** in a congruent direction : positive backgrounds increase & negative backgrounds decrease overall mood ratings

Discussion

- Ensemble emotion perception integrates multiple sources of information** beyond facial expressions: background context, the observer's internal state
- Background emotion biases ensemble judgments**, whereas the **observer's mood modulates response variability**
- These findings highlight **the importance of studying ensemble perception in naturalistic settings** that capture the richness of real social environments

References

Baek, J., & Chong, S. C. (2020). Ensemble perception and focused attention: Two different modes of visual processing to cope with limited capacity. *Psychonomic Bulletin & Review*, 27(4), 602–606.
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Peng, S., Liu, C. H., Liu, W., & Yang, Z. (2022). Emotion matters: Face ensemble perception is affected by emotional states. *Psychonomic Bulletin & Review*, 29(1), 116–122.
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Experiment 2

Autobiographical recall
(10 min)

PANAS (until response)

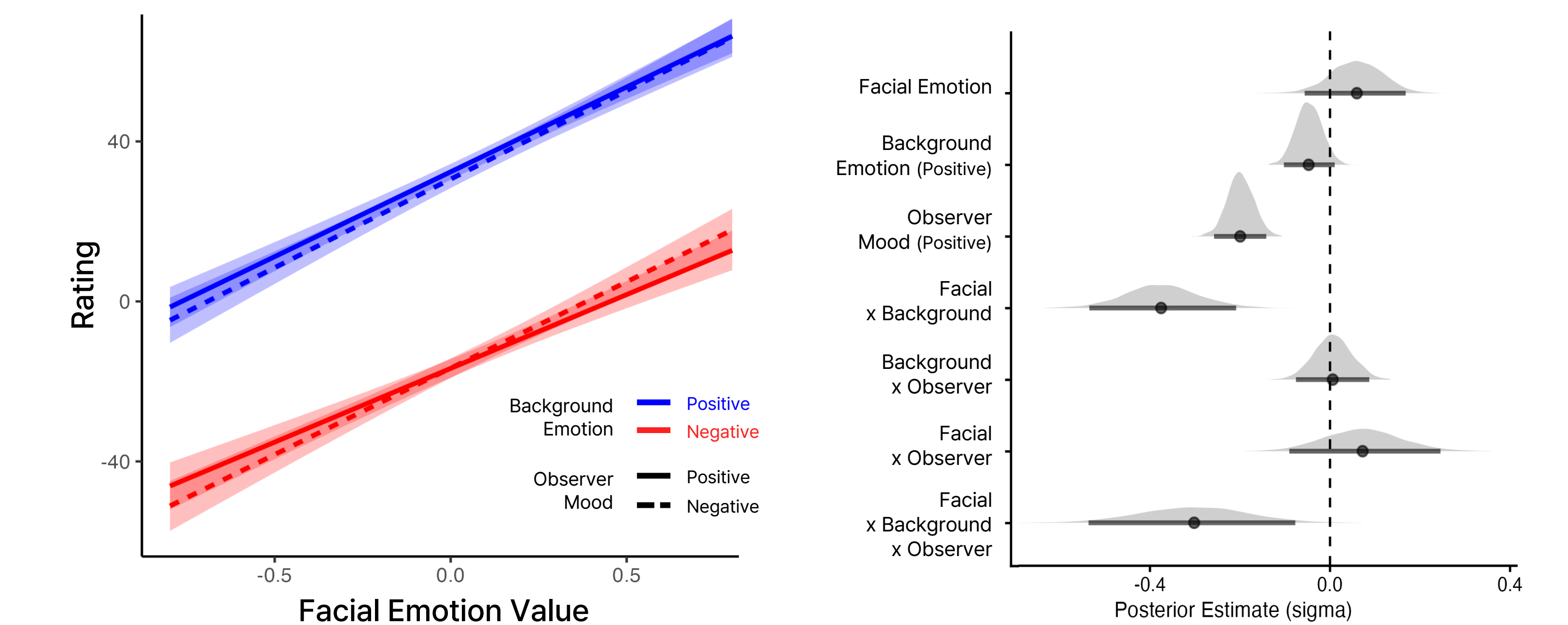
Emotion ensemble task
(Exp 1; 120 trials)

- 2 Observer emotion conditions** : Positive & Negative

- Positive and Negative Affect Scale

- 2 Background emotions** : Positive & Negative

Results (N = 43)



- The **full model** (facial emotion × background emotion × observer mood) provided the **best fit** relative to the 'facial × background interaction model' ($BF_{10} > 10^5$)
- Positive mood reduced response variability**, stabilizing ensemble judgment

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